

IIDS67682 Assignment 2: Transformers and representation learning

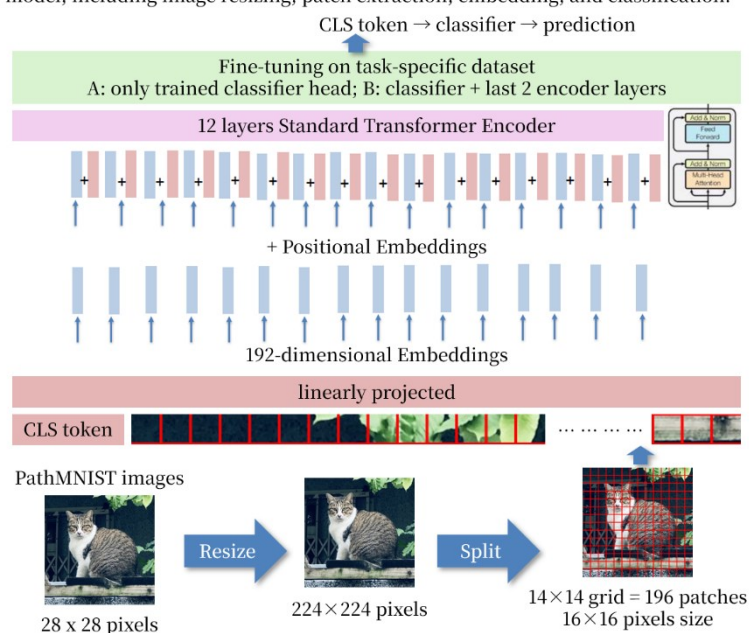
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Methods

The PathMNIST subdataset consists of 26,998 training, 3,001 validation, and 2,154 test images. Each image is a 28×28 RGB histopathological patch derived from haematoxylin and eosin stained tissue sections.

The pre-trained model used was facebook/deit-tiny-patch16-224, a Data-efficient Image Transformer (DeiT), which is an encoder-only Vision Transformer (ViT) trained on the ImageNet-1k dataset (approximately 1 million images across 1,000 classes) at a resolution of 224×224 pixels [1].

Figure 1. Overview of the image processing pipeline for the Vision Transformer (ViT) model, including image resizing, patch extraction, embedding, and classification.



The model processes images (Figure 1) by splitting them into non-overlapping 16×16 patches, which are embedded and combined with positional encodings and a CLS token before being passed through 12 Transformer encoder layers. Each layer consists of multi-head self-attention (3 heads) and a feed-forward network with Gaussian Error Linear Unit (GELU) activation, using pre-layer normalisation. The final CLS representation is used for classification. For this task, the classifier head was replaced with a linear layer for 9 classes. All experiments were conducted with a fixed random seed to ensure reproducibility.

Convolutional Neural Networks (CNNs)

process images through sliding convolutional filters applied to overlapping local patches, building hierarchical spatial representations from low-level features to high-level abstractions. In contrast, ViT divide images into a sequence of non-overlapping fixed-size patches and model global relationships between all patches simultaneously using self-attention. Because ViT do not operate directly on individual pixels but on patch-level embeddings, the attention matrix is computed over a much smaller number of tokens, making the architecture more scalable to higher-resolution inputs than pixel-wise attention would be. However, this global modelling capability typically requires large-scale pre-training data to be effective. Fine-tuning strategies were therefore employed to adapt the pre-trained DeiT model to the smaller PathMNIST dataset without requiring training from scratch.

Two fine-tuning strategies were implemented to evaluate the impact of model adaptability. In Strategy A, all encoder layers were frozen and only the classifier head was trained, effectively using the model as a fixed feature extractor. In Strategy B, the final two encoder layers were unfrozen in addition to the classifier head. Training was performed for 8 epochs using a batch size of 16, learning rate of 2×10^{-5} , and weight decay of 0.01.

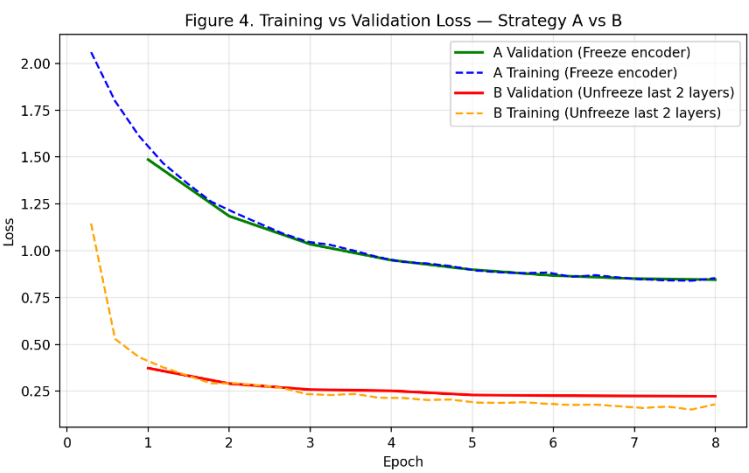
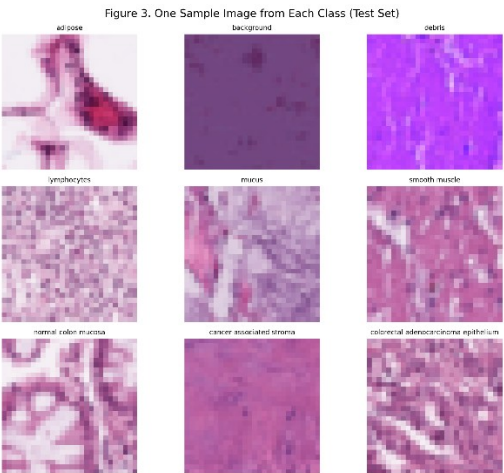
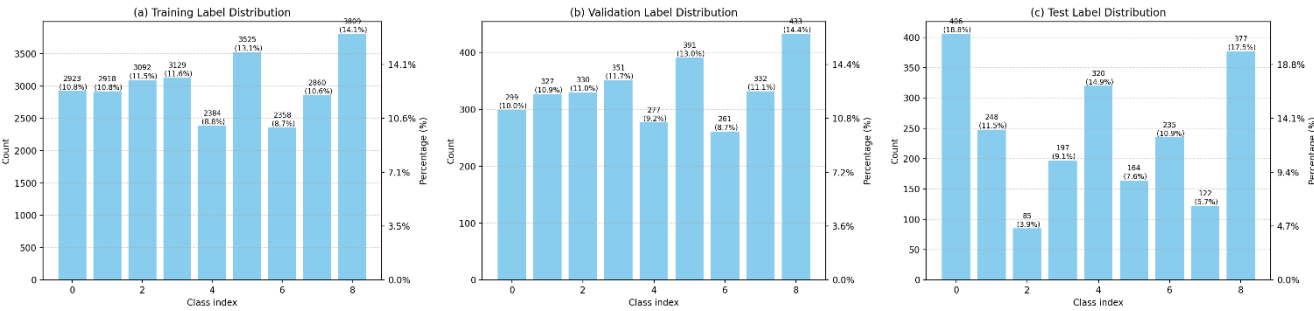
Training and validation loss curves were monitored to assess convergence and potential overfitting. Model performance was evaluated using accuracy, precision, recall, and F1-score. Confusion matrices were used to analyse class-wise errors.

Embeddings were extracted from the CLS token of the final encoder layer for test images. Dimensionality reduction was performed using Principal Component Analysis (PCA), retaining the top two components, and t-distributed Stochastic Neighbour Embedding (t-SNE) with perplexity=30 and random_state=42 for visualisation, and k-means clustering (K = 9, n_init=10, random_state=42) was applied to the original embeddings, and clustering performance was evaluated using the Adjusted Rand Index (ARI), allowing comparison of how well the learned representations reflect true class structure.

Results

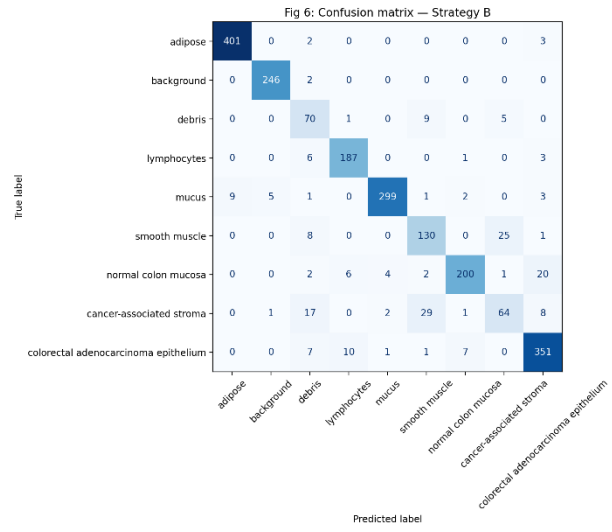
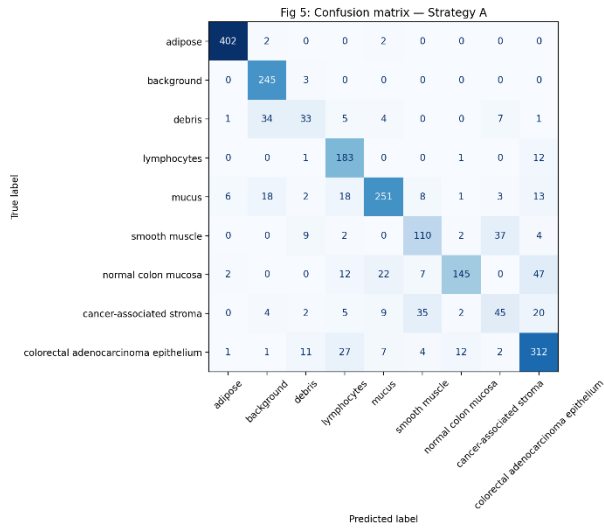
Figure 2 illustrates the label distribution across dataset splits. Training and validation sets show a similar trend: classes mucus and normal colon mucosa have the lowest representation (approximately 9%), classes smooth muscle and colorectal adenocarcinoma epithelium the highest (13–14%), with the remaining classes falling between 10–11%. While the test set shows greater variation (3.9%–18.8%). Figure 3 shows one representative image from each class.

Figure 2. Label Distribution Across Training, Validation, and Test Sets
0: adipose 1: background 2: debris 3: lymphocytes 4: mucus 5: smooth muscle 6: normal colon mucosa 7: cancer-associated stroma 8: colorectal adenocarcinoma epithelium



Training and validation loss curves (Figure 4) show consistent convergence for both strategies. Strategy B achieved a substantially lower final validation loss (0.242) than Strategy A (1.022), indicating improved optimisation when higher-level representations were unfrozen. Strategy B achieved higher test accuracy (0.90) than Strategy A (0.80), indicating that partial unfreezing improved the model’s ability to adapt to the target domain.). Confusion matrices (Figures 5 and 6) show that Strategy A exhibits notable

misclassification in debris (precision=0.54, F1=0.45), smooth muscle (precision=0.67, F1=0.67), and cancer-associated stroma (precision=0.48, F1=0.42), suggesting that feature representations learned from ImageNet are insufficient for distinguishing visually similar histological structures. Strategy B improves performance across all classes, these categories remain comparatively challenging. For example, debris shows improved recall (precision=0.61, F1=0.70), indicating increased true positives at the cost of more false positives. Cancer-associated stroma (precision=0.67, F1=0.59) continues to be confused with smooth muscle, suggesting that these classes share overlapping features that are difficult to separate even after fine-tuning. Overall, the confusion matrices demonstrate that unfreezing the final encoder layers enhances class separability, but does not fully resolve ambiguities between structurally similar tissue types.



Figures 7,8 compare PCA and t-SNE for visualising the learned embeddings. PCA, as a linear dimensionality reduction method, projects the data onto directions of maximum variance. Under Strategy A, this results in substantial overlap between most classes, indicating that the dominant variance directions do not align well with class boundaries. Even under Strategy B, although separation improves, some classes (e.g., cancer-associated stroma and lymphocytes) remain partially overlapping. In contrast, t-SNE, a non-linear technique that preserves local neighbourhood structure, produces more clearly separated clusters for both strategies. This suggests that class-discriminative information is present in the embeddings but is not linearly separable. However, t-SNE also exhibits fragmentation of certain classes (e.g., debris), reflecting its sensitivity to local structure and parameter settings. Therefore, while t-SNE provides better visual separation, PCA offers a more stable global representation of the data. Overall, these results indicate that Strategy B produces embeddings with improved class structure, and that t-SNE is more effective than PCA for visualising non-linear separability in this task.

Figure 7. Only trained classifier head (Strategy A) PCA and t-SNE Projections

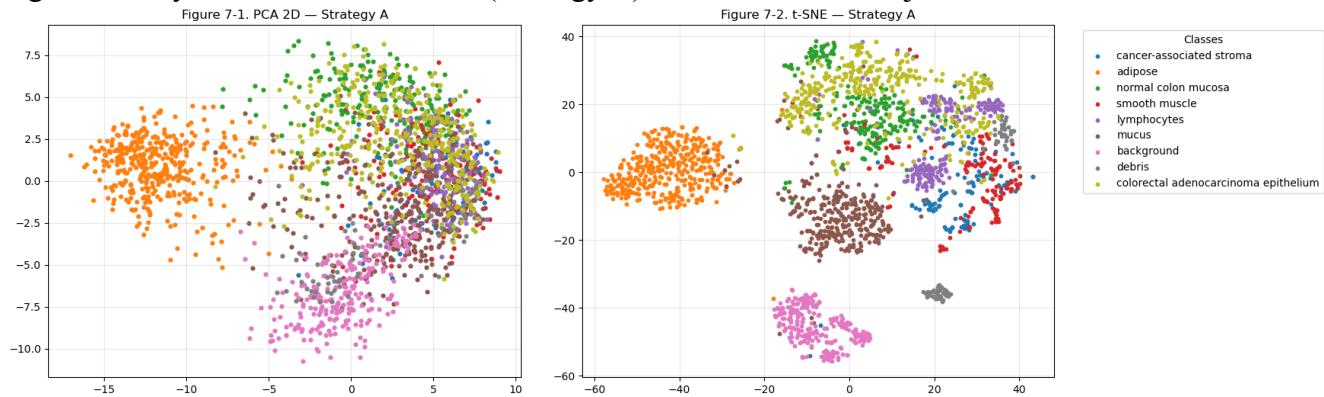
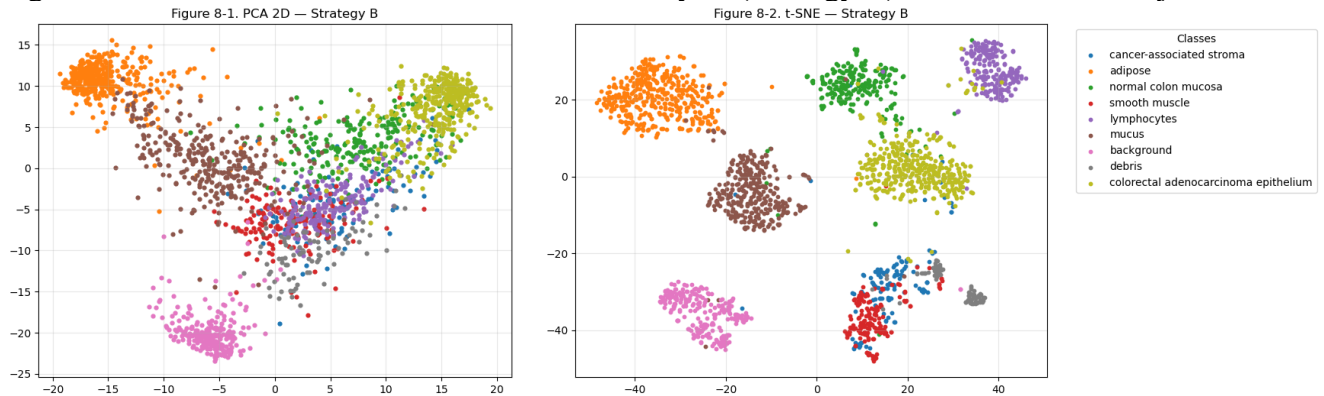


Figure 8. Trained classifier head and unfrozen two layers (Strategy B) PCA and t-SNE Projections



Discussion

In conclusion, Strategy B outperformed Strategy A, attributable to the adaptive fine-tuning of the final two encoder layers on the target dataset. This domain-specific adaptation enabled the model to learn tissue-relevant representations beyond the generic ImageNet features retained by Strategy A, thereby improving image classification across most tissue classes. Nevertheless, cancer-associated stroma, smooth muscle, and debris remained difficult to classify reliably under both strategies. Given the clinical consequences of misclassifying these tissue types in colorectal histology, further investigation would be warranted before deploying such a model in a diagnostic context.

Reference: [1] <https://huggingface.co/facebook/deit-tiny-patch16-224>